

Machine Learning Analysis of Pancreatic Beta-Cell Gene Expression in Type 2 Diabetes Using the GSE20966 Dataset

Abstract

Type 2 diabetes (T2D) is a chronic metabolic disorder where pancreatic beta-cells, which are responsible for producing insulin, gradually lose their normal function. Understanding changes in gene expression within these cells can provide insights into the molecular processes involved in disease development. In this project, a publicly available gene expression dataset (GSE20966) containing pancreatic beta-cell samples from 10 type 2 diabetic and 10 non-diabetic donors was analyzed. The complete analysis was performed using Python-based bioinformatics and machine learning approaches. After preprocessing, cleaning, and scaling the dataset, exploratory analysis was carried out using Principal Component Analysis (PCA) and heatmap visualization to understand overall expression patterns. Differential gene expression analysis was performed to identify genes showing differences between diabetic and control samples. Additionally, three machine learning models—Random Forest, Support Vector Machine (SVM), and Decision Tree, were applied to evaluate their ability to classify samples based on gene expression profiles. The main purpose of this study was to explore diabetes-associated molecular patterns while gaining practical experience in applying data science methods to biological datasets. Due to the limited sample size, the findings are considered exploratory and require further validation using larger datasets.

Introduction

Diabetes is one of the most common chronic diseases worldwide. According to the IDF Diabetes Atlas, around 589 million adults were living with diabetes in 2024, with type 2 diabetes (T2D) accounting for more than 90% of cases (1). In T2D, the body becomes less responsive to insulin, and over time the pancreatic beta-cells are unable to produce enough insulin to maintain normal blood glucose levels (2). Beta-cells, located in pancreatic islets, play an important role in insulin production. During the early stages of diabetes, these cells increase their activity to compensate for insulin resistance, but continuous stress can eventually lead to beta-cell dysfunction (3). Studying changes in beta-cell gene expression can help researchers understand the molecular processes involved in disease development.

Large-scale gene expression studies are available through public databases such as the Gene Expression Omnibus (GEO), allowing researchers to analyze existing biological datasets (4). The dataset used in this project, GSE20966, contains gene expression data from beta-cell samples of 10 type 2 diabetic and 10 non-diabetic donors, generated using microarray technology by Marselli et al. (5).

Machine learning approaches are increasingly being used in biological research to identify patterns within complex datasets. In this project, a Python-based workflow was used to explore the GSE20966 dataset, analyze gene expression patterns, and evaluate whether machine learning models could differentiate between diabetic and control samples.

Objectives

The objectives of this project were:

- To download and prepare the GSE20966 gene expression dataset for analysis in Python.
- To explore the overall structure of the data using PCA and a heatmap.
- To find the genes that show the biggest difference in expression between diabetic and control samples.
- To train and compare three machine learning models — Random Forest, Support Vector Machine and Decision Tree.
- To evaluate how well the models performed and to understand the limitations of working with a small dataset.

Methodology

1. Dataset:

The dataset GSE20966 was obtained from the NCBI Gene Expression Omnibus (5).

Property	Detail
GEO Dataset	GSE20966
Source study	Marselli et al., PLoS One, 2010
Tissue	Pancreatic beta-cell-enriched tissue
Samples	20 total = 10 controls, 10 type 2 diabetic
Labels used	0 = control, 1 = type 2 diabetes

Table 1. Summary of the GSE20966 dataset.

2. Tools used:

Library	What it was used for
pandas, NumPy	Loading and handling the gene expression data
scikit-learn	Scaling, PCA, train/test split, machine learning models
Matplotlib, seaborn	Making the PCA plot, heatmap and confusion matrix

Table 2. Python libraries used in the project.

3.Data Preparation:

The GEO series matrix file was imported into Python using pandas, and only gene expression values were retained. The dataset was transposed so that samples became rows, and genes became columns, making it suitable for machine learning analysis. Samples were labeled as control (0) and diabetic (1), checked for missing values, and standardized using Standard-Scaler.

4.Exploratory Data Analysis:

PCA was used to reduce the high-dimensional gene expression data into two components for visualization. A scatter plot was generated to observe separation between diabetic and control samples. A heatmap of the most variable genes was also used to visualize expression patterns across samples.

5.Gene Expression Comparison:

Mean expression levels were compared between diabetic and control groups to identify genes showing up or down regulation. This provided a simple list of potential candidate genes for further analysis.

6.Machine Learning Classification:

The dataset was split into training (80%) and testing (20%) sets. Random Forest, SVM, and Decision Tree models were trained to classify samples, and their performance was evaluated using accuracy.

Results

The dataset contained 61,295 gene features across 20 samples.

	ID_REF	GSM524151	GSM524152	GSM524153	GSM524154	GSM524155	GSM524156	GSM524157	GSM524158	GSM524159
0	1053_3p_at	6.00	10.08	3.75	3.37	12.06	6.00	5.07	2.41	4.16
1	117_3p_at	48.03	51.81	45.94	44.11	45.94	48.38	44.90	37.28	52.38
2	1494_3p_f_at	57.65	79.37	65.45	61.92	63.53	49.86	75.27	59.12	59.17
3	1552275_3p_s_at	32.35	12.13	17.49	24.62	24.16	20.04	20.95	6.56	5.84
4	1552281_3p_at	51.25	29.51	44.09	35.96	28.19	47.31	45.20	32.01	35.10

Figure 1. Gene expression matrix showing gene identifiers and expression values.

PCA:

PCA was applied to visualize the 20 samples using their first two principal components (Figure 2). The two groups showed a partial separation along the second component, with most diabetic samples located above zero and most control samples below it, though a few samples from each group appeared on the opposite

side. Given the small sample size, this pattern should be interpreted as a general trend rather than a definitive separation.

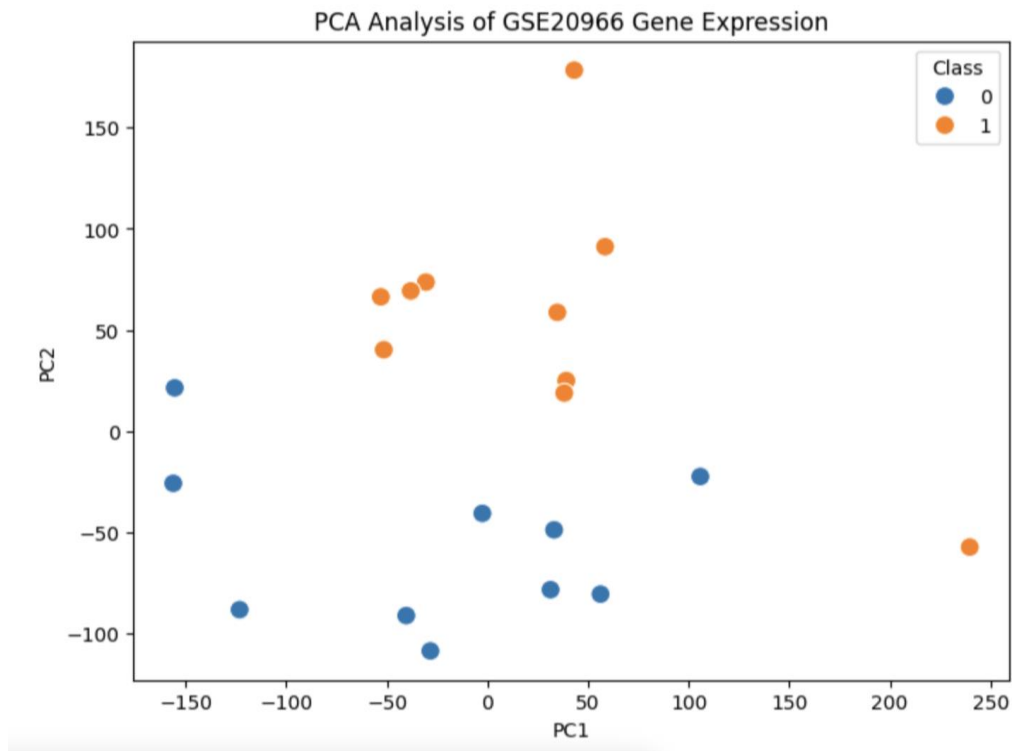


Figure 2. PCA plot of the GSE20966 samples. Most diabetic samples (orange) fall above $PC2 = 0$ while most control samples (blue) fall below it, showing a partial separation between the two groups along the second principal component.

Heatmap (Most variable genes):

The 50 genes that varied the most across the samples were shown as a heatmap (Figure 3). This gives a first look at which genes change the most, though on its own the heatmap does not show an obvious split between the diabetic and control samples.

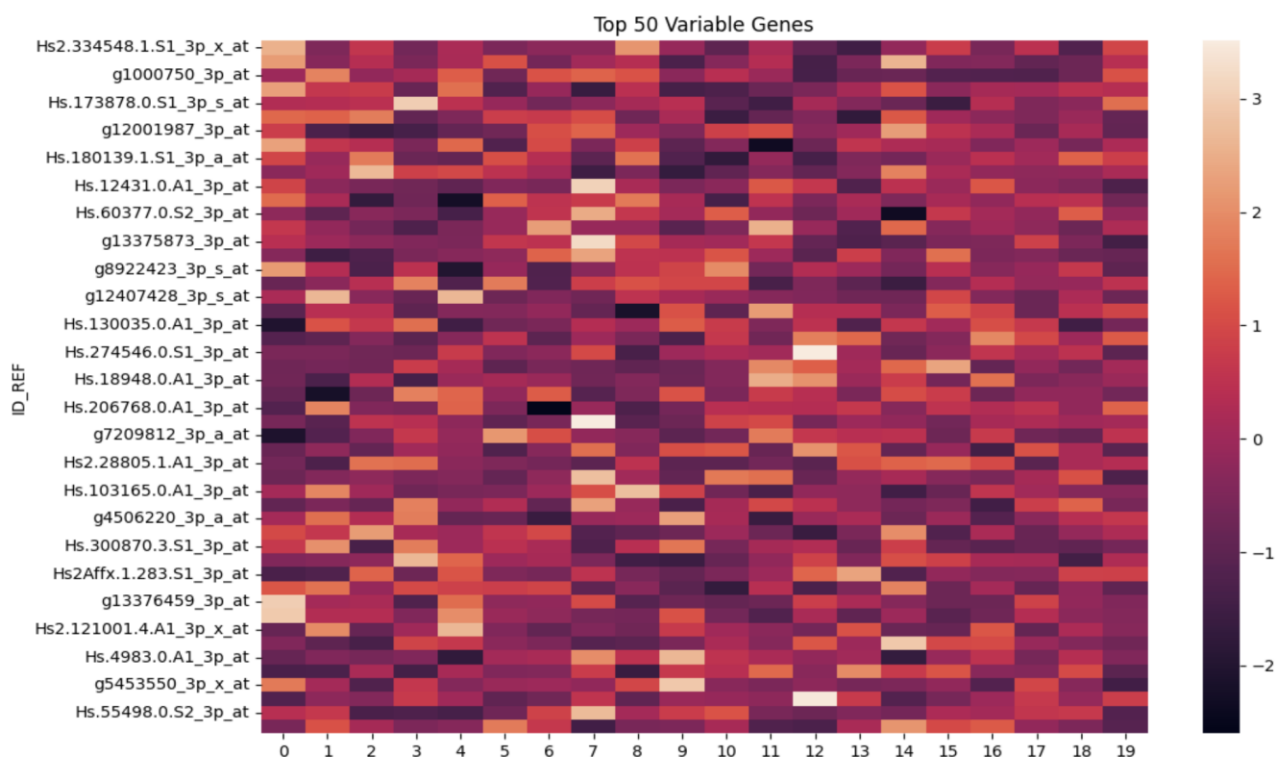


Figure 3. Heatmap of the 50 most variable genes across the samples. Expression varies considerably from sample to sample, but no obvious block pattern separating the two groups is visible by eye.

Genes that changed between the groups:

Sorting the genes by the difference in average expression gave a list of those most increased in the diabetic samples. The top genes from this list are shown in Table 3.

	Probe ID	Values
1	g4505846_3p_s_at	3686.788
2	g11321627_3p_at	3270.650
3	1553970_3p_s_at	2940.001
4	g4502770_3p_s_at	2905.190
5	g13543517_3p_s_at	2888.411

Table 3. Top genes ranked by difference in average expression (diabetic – control).

How the models performed:

The three models were trained on 16 samples and tested on the 4 remaining samples. Their accuracy scores are shown in Table 4. Random Forest and the Support Vector Machine both classified every test sample correctly, while the Decision Tree got only half right, because the test set is very small.

Model	Test accuracy
Random Forest	1.0
Support Vector Machine	1.0
Decision Tree	0.5

Table 4. Accuracy of the three models on the test set.

Discussion & Conclusion

This project demonstrates that a comprehensive gene expression analysis can be successfully carried out on a real dataset using Python and standard data science libraries. By analyzing gene expression data from pancreatic beta-cells, this workflow explored the biological differences between type 2 diabetic and non-diabetic samples through Principal Component Analysis (PCA), heatmaps, differential expression, and machine learning classification.

While the project successfully integrated data cleaning, visualization, and modeling into a single workflow, the findings serve as a preliminary starting point rather than final answers due to a major limitation: dataset size. With only 20 samples in total, the dataset suffers from the "curse of dimensionality," meaning it features thousands of gene features but very few biological replicates. The method used to find differentially expressed genes was a simple fold-change analysis based on group averages, lacking statistical thresholding like p-values. Machine learning models trained on such data risk overfitting—memorizing the training data rather than discovering genuine biological patterns. Consequently, high training performances can be misleading, and slight changes to the data split could heavily swing the results, as evidenced by the high performance of the SVM compared to the much lower score of the Decision Tree on the tiny test set.

Furthermore, the initial method used to identify top genes relied strictly on comparing group averages without evaluating statistical significance. To draw firm biological conclusions and properly compare the findings with established literature, a more robust statistical framework is required. Nevertheless, as a learning exercise, this project successfully demonstrated how computational data science methods can be practically applied to real-world biological questions.

What Could Be Done Better

To improve the reliability and biological relevance of this analysis in future, the following steps should be taken:

- **Utilize larger dataset:** Incorporate more samples from the Gene Expression Omnibus (GEO) to improve model generalization and reduce volatility in performance metrics.
- **Apply robust statistical testing:** Replace simple average comparisons with proper statistical tests (t-tests or ANOVA) to calculate p-values.

References

1. International Diabetes Federation. IDF Diabetes Atlas, 11th edition. Brussels: International Diabetes Federation; 2025.
2. American Diabetes Association Professional Practice Committee. Diagnosis and classification of diabetes: Standards of Care in Diabetes, 2026.
3. Cerf ME. Beta cell dysfunction and insulin resistance. *Frontiers in Endocrinology*. 2013;4:37.
4. Edgar R, Domrachev M, Lash AE. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Research*. 2002
5. Marselli L, Thorne J, Dahiya S, Sgroi DC, Sharma A, Bonner-Weir S, Marchetti P, Weir GC. Gene expression profiles of beta-cell enriched tissue obtained by laser capture microdissection from subjects with type 2 diabetes. *PLoS One*. 2010
6. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*. 2011
7. Ringnér M. What is principal component analysis? *Nature Biotechnology*. 2008;26(3):303–304.
8. Libbrecht MW, Noble WS. Machine learning applications in genetics and genomics. *Nature Reviews Genetics*. 2015;16(6):321–332.
9. Yagin FH, Al-Hashem F, Ahmad I, Ahmad F, Alkhateeb A. Pilot-study to explore metabolic signature of type 2 diabetes: a pipeline of tree-based machine learning and bioinformatics techniques for biomarkers discovery. *Nutrients*. 2024;16(10):1537.
10. Abd El Gwad AM, et al. Machine learning-based stratification of prediabetes and type 2 diabetes progression. *Diabetology & Metabolic Syndrome*. 2025.

